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Fan assembly, fan apparatus, and connector

Abstract

A fan assembly includes two fan apparatuses and at least one connector. Each of the two fan apparatuses includes a frame, a fan blade and at least one first terminal. The frame has at least one first engagement recess. The fan blade is rotatably disposed on the frame. The at least one first terminal is located in the at least one first engagement recess. The at least one connector includes a first assembly block and two second terminals. The first assembly block includes two first engaging portions connected to each other. The two second terminals are respectively mounted on the two first engaging portions. The two first engaging portions of the first assembly block are respectively engaged with the at least two first engagement recesses of the two frames, and the two second terminals are respectively plugged and electrically connected to the at least two first terminals.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No(s). 202211222135.3 filed in China, on Oct. 8, 2022, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a fan assembly, more particularly to a fan assembly integrating

the installation and series circuit connection of two fans.
BACKGROUND
(3) With the rapid development of technology, the computing performance of various electronic components has also increased significantly, and a large amount of heat has been generated at the same time. To ensure that the electronic components won't be damaged due to high temperature, it is necessary to install a cooling apparatus on an electronic product to dissipate the excessive heat of the electronic components. A common cooling apparatus is a fan, which drives air to flow through the electronic components and remove heat via the operation of the fan so that the electronic components can operate within a certain operating temperature range. Sometimes the cooling capacity of a single fan may not be sufficient, so two or more fans are usually connected in series to form a fan wall to provide sufficient air volume
(4) However, currently, when the plurality of fans are to be connected in series, in addition to fixing the fans via screws or snap-fit elements for a mechanical connection, it is necessary that a plurality of cables are respectively plugged to the fans for a series circuit connection. When the plurality of fans are arranged side by side, the space between the fans is limited, which further increases the difficulty for operators to lock screws or plug cables. Therefore, how to simplify the serial connection procedure of the plurality of fans is one of the problems that developers should address.
SUMMARY
(5) The present disclosure provides a fan assembly capable of simplifying the serial connection procedure of two fans.
(6) One embodiment of the present disclosure provides a fan assembly including two fan apparatuses and at least one connector. Each of the two fan apparatuses includes a frame, a fan blade and at least one first terminal. The frame has at least one first engagement recess. The fan blade is rotatably disposed on the frame. The at least one first terminal is located in the at least one first engagement recess. The at least one connector includes a first assembly block and two second terminals. The first assembly block includes two first engaging portions connected to each other. The two second terminals are respectively mounted on the two first engaging portions. The two first engaging portions of the first assembly block are respectively engaged with the at least two first engagement recesses of the two frames, and the two second terminals are respectively plugged and electrically connected to the at least two first terminals.
(7) Another embodiment of the present disclosure provides a fan apparatus including a frame, a fan blade and at least one first terminal. The frame has at least one first engagement recess. The fan blade is rotatably disposed on the frame. The at least one first terminal is located in the at least one first engagement recess.
(8) Another embodiment of the present disclosure provides a connector configured to connect a plurality of fan apparatuses. The connector includes a first assembly block and two terminals. The first assembly block includes two first engaging portions connected to each other, and the first assembly block has at least one first movement limiting recess. The two terminals is respectively mounted on the two first engaging portions.
(9) According to the fan assembly, fan apparatus and connector as described above, since the connector also includes an engaging portion for a mechanical connection and a terminal for a series circuit connection, the operator can complete the mechanical connection and series circuit connection of the two fan apparatuses at one time via the assembly of the connector and the two fan apparatuses so as to integrate the mechanical connection of the two fans and the series connection of the circuit, thereby simplifying the procedure of connecting the two fans in series.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only and thus are not limitative of the present disclosure and wherein:

(2) FIG. **1** is a perspective view of a fan assembly in accordance with the first embodiment of the present disclosure;

(3) FIG. **2** is a partially exploded view of the fan assembly in FIG. **1**;

(4) FIG. **3** is a perspective view of a fan apparatus in accordance with the first embodiment of the present disclosure;

(5) FIG. **4** is a perspective view of a connector in accordance with the first embodiment of the present disclosure; and

(6) FIG. **5** is a perspective view of a fan assembly in accordance with the second embodiment of the present disclosure.

DETAILED DESCRIPTION

(7) In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

(8) In addition, the terms used in the present disclosure, such as technical and scientific terms, have its own meanings and can be comprehended by those skilled in the art, unless the terms are additionally defined in the present disclosure. That is, the terms used in the following paragraphs should be read on the meaning commonly used in the related fields and will not be overly explained, unless the terms have a specific meaning in the present disclosure.

(9) Referring to FIG. **1** and FIG. **2**, FIG. **1** is a perspective view of a fan assembly in accordance with the first embodiment of the present disclosure, and FIG. **2** is a partially exploded view of the fan assembly in FIG. **1**.

(10) In this embodiment, a fan assembly **10** is provided. The fan assembly **10** includes two fan apparatuses **11** and a connector **12**. Each of the two fan apparatuses **11** includes a frame **111**, a fan blade **112**, two first terminals **113**, and two first protruding portions **114**. The frame **111** has an inner annular surface A**1**, an outer annular surface A**2** and an annular front surface A**3**. The inner annular surface A**1** surrounds a fan blade accommodation space S, and the outer annular surface A**2** surrounds the fan blade **112** and faces away from the inner annular surface A**1**. The annular front surface A**3** connects with the outer annular surface A**2**. The frame **111** has two first engagement recesses F**1**. The two first engagement recesses F**1** are recessed inward from the outer annular surface A**2** to form two first openings O**1**. The two first engagement recesses F**1** connect with the annular front surface A**3**. The two first protruding portions **114** are disposed on the frame **111** and cover a part of the two first openings O**1**. The two first terminals **113** are located in the two first engagement recesses F**1**. The fan blade **112** is rotatably disposed in the fan blade accommodation space S.

(11) The connector **12** includes a first assembly block **121** and two second terminals **122**. The first assembly block **121** includes two first engaging portions **1211** connected to each other. The two second terminals **122** are respectively mounted on the two first engaging portions **1211**. The two first engaging portions **1211** of the first assembly block **121** are respectively engaged with the two first engagement recesses F**1** of the two frames **111**, and the two second terminals **122** are respectively plugged and electrically connected to the two first terminals **113**. The first assembly block **121** has a first movement limiting recess L**1**. The two first protruding portions **114** of the two frames **111** are located in the first movement limiting recess L**1** to limit a relative movement of the two fan apparatuses **11**. Accordingly, the connector **12** is connected to the two frames **111**, and the

first movement limiting recess L**1** further limits the relative movement of the two fan apparatuses **11**. In addition, the connector **12** is electrically connected to the two fan apparatuses **11** via the two first terminals **113** and the two second terminals **122**, so the connector **12** is also connected to the circuit of each of the fan apparatuses **11** in the fan assembly **10** in series at the same time. That is, only one of the first terminals **113** is electrically connected to a power supply unit, and the power supply unit can drive all of the fan apparatuses **11** of the fan assembly **10** to operate.

(12) In this embodiment, the quantity of the fan apparatuses **11** of the fan assembly **10** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the fan apparatus of the fan assembly may be more than two.

(13) In this embodiment, the quantity of the connector **12** is one, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the connector may be more than one.

(14) In this embodiment, the quantity of the first terminal **113** of each of the fan apparatuses **11** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first terminal may be only one.

(15) In this embodiment, the quantity of the first protruding portion **114** of each of the fan apparatuses **11** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first protruding portion may be only one.

(16) In this embodiment, the two first protruding portion **114** and the frame **111** are formed of a single piece, but the present disclosure is not limited thereto. In some other embodiments, the two first protruding portion may be independent of the frame and disposed on the frame.

(17) In this embodiment, the quantities of the first engagement recess F**1** and the first opening O**1** of each of the fan apparatuses **11** are two, but the present disclosure is not limited thereto. In some other embodiments, the quantities of the first engagement recess and the first opening may be only one.

(18) In this embodiment, the two first engagement recesses F**1** are recessed inward from the outer annular surface A**2** to form the two first openings O**1**, but the present disclosure is not limited thereto. In some other embodiments, the two first engagement recesses may be located between the outer annular surface and the inner annular surface, and the first engagement recesses are separated from the outer annular surface and the inner annular surface.

(19) In this embodiment, the quantity of the first movement limiting recess L**1** of the connector **12** is one, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first movement limiting recess may be more than one.

(20) In this embodiment, the purpose of providing the second terminals **122** in the connector **12** is that in addition to connecting the two fan apparatuses **11** via the engaging portion and the movement limiting recess, the circuit of each of the fan apparatuses **11** is connected in series via an electrical connection between the first terminal **113** of the fan apparatus **11** and the second terminal **122** of the connector **12**. Therefore, via the design of providing the terminals respectively on the engaging portions of the connector **12** and the engaging recesses of the fan apparatus **11**, the structural assembly and series connection of circuits of two fans can be integrated, thereby simplifying the serial connection procedure of the fans.

(21) In this embodiment, each of the two fan apparatuses **11** further includes two second protruding portions **115**. The frame **111** of each of the fan apparatuses **11** further has two second engagement recesses F**2** and two accommodation recesses C. The two second engagement recesses F**2** and the two first engagement recesses F**1** are located on the same side with respect to the fan blade **112**. The two second engagement recesses F**2** are recessed inward from the outer annular surface A**2** to form two second openings O**2**. The two second protruding portions **115** are disposed on the frame **111** and covers a part of the two second openings O**2**.

(22) In this embodiment, the connector **12** further includes a second assembly block **123** and a connecting component **124**. The connecting component **124** is connected to the first assembly block **121** and the second assembly block **123**. The second assembly block **123** includes two

second engaging portions **1231** connected to each other. The two second engaging portions **1231** of the second assembly block **123** are respectively engaged with the two second engagement recesses F**2** of the two frames **111**. The second assembly block **123** has a second movement limiting recess L**2**. The two second protruding portions **115** of the two frames **111** are located in the second movement limiting recess L**2** to limit the relative movement of the two fan apparatuses **11**. The two accommodation recesses C are located at the outer annular surface A**2** and between the two first engagement recesses F**1** and the two second engagement recesses F**2** respectively, and the two accommodation recesses C are recessed inward from the outer annular surface A**2**. The two accommodation recesses C are configured to accommodate the connecting component **124** of the connector **12**. Accordingly, the first assembly block **121** and the second assembly block **123** of the connector **12** connect the two frames **111**, and the first movement limiting recess L**1** and the second movement limiting recess L**2** further restrict the relative movement of the two fan apparatuses **11**.
(23) In this embodiment, the quantity of the second protruding portion **115** of each of the fan apparatuses **11** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the second protruding portion may be only one.
(24) In this embodiment, the quantities of the second engagement recess F**2** and the second opening O**2** of each of the fan apparatuses **11** are two, but the present disclosure is not limited thereto. In some other embodiments, the quantities of the second engagement recess and the second opening may be only one.
(25) In this embodiment, the two second engagement recesses F**2** are recessed inward from the outer annular surface A**2** to form the two second openings O**2**, but the present disclosure is not limited thereto. In some other embodiments, the two second engagement recesses may be located between the outer annular surface and the inner annular surface, and the second engagement recesses are separated from the outer annular surface and the inner annular surface.
(26) In this embodiment, the quantity of the accommodation recess C of each of the fan apparatuses **11** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the accommodation recess may be only one.
(27) In this embodiment, the quantity of the second movement limiting recess L**2** of the connector **12** is one, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the second movement limiting recess may be more than one.
(28) In this embodiment, the fan assembly **10** may further include an external wire **13** and an external terminal **131**. The external wire **13** is electrically connected to a power supply unit. The external terminal **131** is disposed on the external wire **13**. The external terminal **131** is plugged and electrically connected to one of the first terminals **113**. Therefore, the power supply unit can transmit power via the external wire **13** and drive the fan assembly **10** to operate.
(29) Referring to FIG. **3**, FIG. **3** is a perspective view of a fan apparatus in accordance with the first embodiment of the present disclosure.
(30) In this embodiment, the fan apparatus **11** includes a frame **111**, a fan blade **112**, two first terminals **113**, and two first protruding portions **114**. The frame **111** has an inner annular surface A**1**, an outer annular surface A**2** and an annular front surface A**3**. The inner annular surface A**1** surrounds a fan blade accommodation space S, and the outer annular surface A**2** surrounds the fan blade **112** and faces away from the inner annular surface A**1**. The annular front surface A**3** connects with the outer annular surface A**2**. The frame **111** has two first engagement recesses F**1**. The two first engagement recesses F**1** are recessed inward from the outer annular surface A**2** to form two first openings O**1**. The two first engagement recesses F**1** connect with the annular front surface A**3**. The two first terminals **113** are located in the two first engagement recesses F**1**. The fan blade is rotatably disposed on the fan blade accommodation space S. Accordingly, the plurality of the fan apparatuses **11** can be engaged with the two first engagement recesses F**1** and a connector, and the two first protruding portions **114** can be located in a movement limiting recess of the connector. The movement limiting recess can limit the relative movement of the two adjacent fan apparatuses

11. In addition, the first terminal **113** can be electrically connected to the connector including a terminal, so the connector can be also connected to the circuits of each of the fan apparatuses **11** in series at the same time. That is, only one of the first terminals **113** of the plurality of the fan apparatuses **11** is electrically connected to a power supply unit, and the power supply unit can drive each of the fan apparatuses **11** to operate.
(31) In this embodiment, the quantity of the first terminal **113** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first terminal may be only one.
(32) In this embodiment, the quantity of the first protruding portion **114** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first protruding portion may be only one.
(33) In this embodiment, the quantities of the first engagement recess F**1** and the first opening O**1** are two, but the present disclosure is not limited thereto. In some other embodiments, the quantities of the first engagement recess and the first opening may be only one.
(34) In this embodiment, the two first engagement recesses F**1** are recessed inward from the outer annular surface A**2** to form the two first openings O**1**, but the present disclosure is not limited thereto. In some other embodiments, the two first engagement recesses may be located between the outer annular surface and the inner annular surface, and the first engagement recesses are separated from the outer annular surface and the inner annular surface.
(35) In this embodiment, the fan apparatus **11** further includes two second protruding portions **115**. The frame **111** of the fan apparatus **11** further has two second movement limiting recesses F**2** and two accommodation recesses C. The two second engagement recesses F**2** and the two first engagement recesses F**1** are located on a same side relative to the fan blade **112**. The two second engagement recesses F**2** are recessed inward from the outer annular surface A**2** to form the two second openings O**2**. The two second protruding portions **115** are disposed on the frame **111** and covers a part of the two second openings O**2**. The two accommodation recesses C are located at the outer annular surface A**2** and between the two first engagement recesses F**1** and the two second engagement recesses F**2** respectively, and the two accommodation recesses C are recessed inward from the outer annular surface A**2**. Accordingly, the plurality of the fan apparatuses **11** can be engaged with a connector via the first engagement recesses F**1** and the second engagement recesses F**2**, and the first protruding portions **114** and the second protruding portions **115** can be respectively located in two movement limiting recesses of the connector. The two movement limiting recesses can limit the relative movement of the two adjacent fan apparatuses **11**.
(36) In this embodiment, the quantity of the second protruding portion **115** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the second protruding portion may be only one.
(37) In this embodiment, the quantities of the second engagement recess F**2** and the second opening O**2** are two, but the present disclosure is not limited thereto. In some other embodiments, the quantities of the second engagement recess and the second opening may be only one.
(38) In this embodiment, the two second engagement recesses F**2** are recessed inward from the outer annular surface A**2** to form the two second openings O**2**, but the present disclosure is not limited thereto. In some other embodiments, the two second engagement recesses may be located between the outer annular surface and the inner annular surface, and the second engagement recesses are separated from the outer annular surface and the inner annular surface.
(39) In this embodiment, the quantity of the accommodation recess C is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the accommodation recess may be only one.
(40) Referring to FIG. **4**, which is a perspective view of a connector in accordance with the first embodiment of the present disclosure.
(41) In this embodiment, a connector **12** includes a first assembly block **121** and two terminals **122**. The terminal **122** is the second terminal **122** described in the first embodiment. The first assembly

block **121** includes two first engaging portions **1211** connected to each other, and the first assembly block **121** has a first movement limiting recess L**1**. The two terminals **122** are respectively mounted on the two first engaging portions **1211**. Accordingly, the first assembly block **121** of the connector **12** can be connected to the two fan apparatuses, and the first movement limiting recess L**1** can engaged with the protruding portions of the two fan apparatuses to further restrict the relative movement of the two fan apparatuses. In addition, the two terminals **122** of the connector **12** can be further electrically connected to the two fan apparatuses so that the power can drive the two fan apparatuses to operate at the same time.
(42) In this embodiment, the quantity of the first movement limiting recess L**1** is one, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first movement limiting recess may be more than one.
(43) In this embodiment, the connector **12** further includes a second assembly block **123** and a connecting component **124**. The connecting component **124** is connected to the first assembly block **121** and the second assembly block **123**. The second assembly block **123** includes two second engaging portions **1231** connected to each other. The two second engaging portions **1231** of the second assembly block **123** are respectively engaged with the two second engagement recesses F**2** of the two frames **111**. The second assembly block **123** has a second movement limiting recess L**2**. Accordingly, the first assembly block **121** and the second assembly block **123** of the connector **12** can be connected to the two fan apparatuses, and the first movement limiting recess L**1** and the second movement limiting recess L**2** can engaged with the protruding portions of the two fan apparatuses to further restrict the relative movement of the two fan apparatuses.
(44) In this embodiment, the quantity of the second movement limiting recess L**2** is one, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the second movement limiting recess may be more than one.
(45) Referring to FIG. **5**, FIG. **5** is a perspective view of a fan assembly in accordance with the second embodiment of the present disclosure.
(46) In this embodiment, a fan assembly **10**A is provided. The fan assembly **10**A includes two fan apparatuses **11** and a connector **12**. Each of the two fan apparatuses **11** includes a frame **111**, a fan blade **112**, two first terminals **113**, and two first protruding portions **114**. The frame **111** has an inner annular surface A**1**, an outer annular surface A**2** and an annular front surface A**3**. The inner annular surface A**1** surrounds a fan blade accommodation space S, and the outer annular surface A**2** surrounds the fan blade **112** and faces away from the inner annular surface A**1**. The annular front surface A**3** connects with the outer annular surface A**2**. The frame **111** has two first engagement recesses F**1**. The two first engagement recesses F**1** are recessed inward from the outer annular surface A**2** to form two first openings O**1**. The two first engagement recesses F**1** connect with the annular front surface A**3**. The two first protruding portions **114** are disposed on the frame **111** and covers a part of the two first openings O**1**. The two first terminals **113** are located in the two first engagement recesses F**1**. The fan blade is rotatably disposed on the fan blade accommodation space S.
(47) The connector **12** includes a first assembly block **121** and two second terminals **122**. The first assembly block **121** includes two first engaging portions **1211** connected to each other. The two second terminals **122** are respectively mounted on the two first engaging portions **1211**. The two first engaging portions **1211** of the first assembly block **121** are respectively engaged with the two first engagement recesses F**1** of the two frames **111**, and the two second terminals **122** are respectively plugged and electrically connected to the two first terminals **113**. The first assembly block **121** has a first movement limiting recess L**1**. The two first protruding portions **114** of the two frames **111** are located in the first movement limiting recess L**1** to limit the relative movement of the two fan apparatuses **11**. Accordingly, the connector **12** is connected to the two frames **111**, and the first movement limiting recess L**1** further limits the relative movement of the two fan apparatuses **11**. In addition, the connector **12** is electrically connected to the two fan apparatuses **11**

via the two first terminals **113** and the two second terminals **122**, so the connector **12** is also connected to the circuit of each of the fan apparatuses **11** in the fan assembly **10** in series at the same time. That is, only one of the first terminals **113** is electrically connected to a power supply unit, and the power supply unit can drive each of the fan apparatuses **11** of the fan assembly **10**A to operate.

(48) In this embodiment, the quantity of the fan apparatuses **11** of the fan assembly **10**A is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the fan apparatus of the fan assembly may be more than two.

(49) In this embodiment, the quantity of the connector **12** is one, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the connector may be more than one.

(50) In this embodiment, the quantity of the first terminal **113** of each of the fan apparatuses **11** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first terminal may be only one.

(51) In this embodiment, the quantity of the first protruding portion **114** of each of the fan apparatuses **11** is two, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first protruding portion may be only one.

(52) In this embodiment, the two first protruding portion **114** and the frame **111** are formed integrally, but the present disclosure is not limited thereto. In some other embodiments, the two first protruding portion may be independent of the frame and set on the frame.

(53) In this embodiment, the quantities of the first engagement recess F**1** and the first opening O**1** of each of the fan apparatuses **11** are two, but the present disclosure is not limited thereto. In some other embodiments, the quantities of the first engagement recess and the first opening may be only one.

(54) In this embodiment, the two first engagement recesses F**1** are recessed inward from the outer annular surface A**2** to form the two first openings O**1**, but the present disclosure is not limited thereto. In some other embodiments, the two first engagement recesses may be located between the outer annular surface and the inner annular surface, and the first engagement recesses are separated from the outer annular surface and the inner annular surface.

(55) In this embodiment, the quantity of the first movement limiting recess L**1** of the connector **12** is one, but the present disclosure is not limited thereto. In some other embodiments, the quantity of the first movement limiting recess may be more than one.

(56) In this embodiment, the purpose of disposing the second terminals **122** in the connector **12** is that in addition to connecting the two fan apparatuses **11** via the engaging portion and the movement limiting recess, the circuit of each of the fan apparatuses **11** is connected in series via an electrical connection between the first terminal **113** of the fan apparatus **11** and the second terminal **122** of the connector **12**. Therefore, via the design of disposing the terminals respectively on the engaging portions of the connector **12** and the engaging recesses of the fan apparatus **11**, the assembly of two fans and series connection of circuits can be integrated, thereby simplifying the serial connection procedure of the fans.

(57) In this embodiment, the fan assembly **10**A may further include an external wire **13** and an external terminal **131**. The external wire **13** is electrically connected to a power supply unit. The external terminal **131** is disposed on the external wire **13**. The external terminal **131** is plugged and electrically connected to one of the first terminals **113**. Therefore, the power supply unit can transmit power via the external wire **13** and drive the fan assembly **10**A to operate at the same time.

(58) According to the fan assembly, fan apparatus and connector as described above, since the connector includes an engaging portion for mechanical assembly and a terminal for series circuit connection, the operator can complete the mechanical connection and series circuit connection of the two fan apparatuses at one time via the assembly of the connector and the two fan apparatuses so as to integrate the mechanical connection and the series connection of the circuits of the two fans, thereby simplifying the procedure of connecting the two fans in series.

(59) It will be apparent to those skilled in the art that various modifications and variations can be made to the present disclosure. It is intended that the specification and examples be considered as exemplary embodiments only, with a scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A fan assembly, comprising: two fan apparatuses, each of the two fan apparatuses comprising: a frame, having at least one first engagement recess; a fan blade, rotatably disposed on the frame; and at least one first terminal, wherein each of the respective one of the at least one first terminal is located in a corresponding one of the at least one first engagement recess; and at least one connector, each of the at least one connector comprising: a first assembly block, wherein the first assembly block comprises two first engaging portions connected to each other; and two second terminals, wherein each of the two second terminals is respectively mounted on a corresponding one of the two first engaging portions, each of the two first engaging portions is respectively engaged with a corresponding one of the at least one first engagement recess, and each of the two second terminals are respectively plugged and electrically connected to a corresponding one of the at least one first terminal; wherein each of the two frames of the two fan apparatuses has an outer annular surface and an annular front surface; in each of the two fan apparatuses, the outer annular surface surrounds the fan blade, the annular front surface connects with the outer annular surface, and the at least one first engagement recess is recessed inward from the outer annular surface to form a first opening and the at least one first engagement recess connects with the annular front surface.
2. The fan assembly according to claim 1, wherein each of the two fan apparatuses further comprises at least one first protruding portion, in each of the two fan apparatuses, each of the at least one first protruding portion is disposed on the frame and covers a part of a corresponding one of the first opening.
3. The fan assembly according to claim 2, wherein first assembly blocks has at least one first movement limiting recess, and each of the at least one first protruding portion is located in a corresponding one of the at least one first movement limiting recess to limit a relative movement of the two fan apparatuses.
4. The fan assembly according to claim 3, wherein each of the two frames of the two fan apparatuses has at least one second engagement recess, in each of the two fan apparatuses, one of the at least one second engagement recess and one of the at least one first engagement recess are located on a same side relative to the fan blade, and the at least one second engagement recess is recessed inward from the outer annular surface to form a second opening; each of the at least one connector further comprises a second assembly block and a connecting component; in each of the at least one connector, the connecting component is connected to the first assembly block and the second assembly block, the second assembly block comprises two second engaging portions connected to each other, and each of the two second engaging portions of the second assembly block are respectively engaged with a corresponding one of the at least one two-second engagement recess recesses of the two frames.
5. The fan assembly according to claim 4, wherein each of the two fan apparatuses further comprises at least one second protruding portion; in each of the two fan apparatuses, the at least one second protruding portion is disposed on the frame and covers a part of the second opening.
6. The fan assembly according to claim 5, wherein the second assembly blocks has at least one second movement limiting recess, and each of the at least one second protruding portion is located in a corresponding one of the at least one second movement limiting recess to limit the relative movement of the two fan apparatuses.
7. The fan assembly according to claim 6, wherein each of the two frames of the two fan

apparatuses has at least one accommodation recess; in each of the two fan apparatuses, the at least one accommodation recess is located at the outer annular surface and between the one of the at least one first engagement recess and the one of the at least one second engagement recess that are located on the same side relative to the fan blade, and the at least one accommodation recess is recessed inward from the outer annular surface; each of the at least one accommodation recess of the two frames that are corresponding to each other is configured to accommodate the connecting component of the at least one connector.

8. A fan apparatus, comprising: a frame, having at least one first engagement recess; a fan blade, rotatably disposed on the frame; and at least one first terminal, wherein each of the at least one first terminal is located in a corresponding one of the at least one first engagement recess; wherein the frame has an outer annular surface and an annular front surface, the outer annular surface surrounds the fan blade, the annular front surface connects with the outer annular surface, and the at least one first engagement recess is recessed inward from the outer annular surface to form a first opening and the at least one first engagement recess connects with the annular front surface.

9. The fan apparatus according to claim 8, wherein the fan apparatus further comprises at least one first protruding portion, and each of the at least one first protruding portion is disposed on the frame and covers a part of the first opening.

10. The fan apparatus according to claim 9, wherein the frame of the fan apparatus has at least one second engagement recess, one of the at least one second engagement recess and one of the at least one first engagement recess are located on a same side relative to the fan blade, and the one of the at least one second engagement recess is recessed inward from the outer annular surface to form a second opening.

11. The fan apparatus according to claim 10, wherein the fan apparatus further comprises at least one second protruding portion, and each of the at least one second protruding portion is disposed on the frame and covers at least a part of the second opening.

12. The fan apparatus according to claim 11, wherein the frame of the fan apparatus has at least one accommodation recess, the at least one accommodation recess is located at the outer annular surface and between the one of the at least one first engagement recess and the one of the at least one second engagement recess that are located on the same side relative to the fan blade, and the one of the at least one accommodation recess is recessed inward from the outer annular surface.